# Practice Paper 2

**MATHEMATICS** **Syllabus 9709**

Paper 3 Pure Mathematics 3 **Written in the 2026 style**

**1 hour 50 minutes**

You must answer on separate answer paper.

You will need: Answer paper
List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Write your answers on the answer paper. Write the question number next to each answer.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

*This practice paper was written in the style of the 2026 question papers. Most questions start with routine parts. Some later parts join two different topics, use general constants, or give a condition in place of an instruction. It is not an official Cambridge International document.*

This document has 6 pages.

**[Turn over**

**1** Solve the equation $2\ln x - \ln(x - a) = \ln(4a)$, where $a$ is a positive constant. Give your answer in terms of $a$. [4]

**2** It is given that the expansion of

$$\frac{1 + ax}{\sqrt{1 + bx}}$$

in ascending powers of $x$, up to and including the term in $x^2$, is $1 + 2x^2$, where $a$ and $b$ are positive constants.

**(a)** Find the values of $a$ and $b$. [4]

**(b)** State the set of values of $x$ for which the expansion is valid. [1]

**3** **(a)** Express $3\cos x + 4\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$. Give the exact value of $R$ and the value of $\alpha$ correct to 4 significant figures. [3]

**(b)** Find the set of values of the constant $k$ for which the equation $3\cos x + 4\sin x = k$ has exactly two roots for $0 \leqslant x \leqslant \pi$. [3]

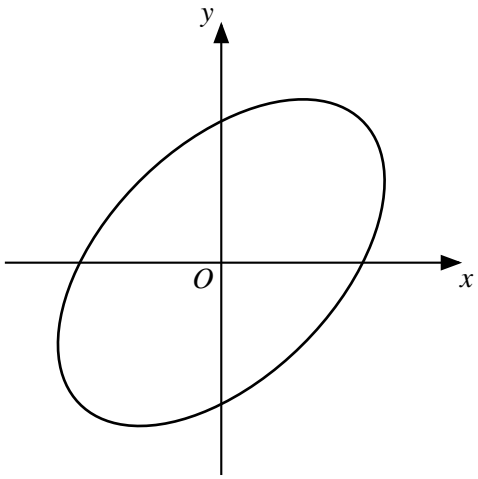
**4** **(a)** Find the square roots of $-15 + 8\mathrm{i}$. Give your answers in the form $x + \mathrm{i}y$, where $x$ and $y$ are real. [4]

**(b)** Hence, or otherwise, solve the equation

$$z^2 - (3 + 2\mathrm{i})z + 5 + \mathrm{i} = 0.$$

Give your answers in the form $x + \mathrm{i}y$. [3]

**5**




The diagram shows the curve with equation $x^2 - xy + y^2 = 12$.

**(a)** Show that $\frac{dy}{dx} = \frac{2x - y}{x - 2y}$. [3]

**(b)** Find the coordinates of the points on the curve at which the tangent is parallel to the $x$-axis. [2]

**(c)** The line $x + y = k$, where $k$ is a positive constant, is a tangent to the curve.

Find the exact value of $k$. [4]

**6** Let

$$f(x) = \frac{4x}{(x + 1)(x^2 + 1)}.$$

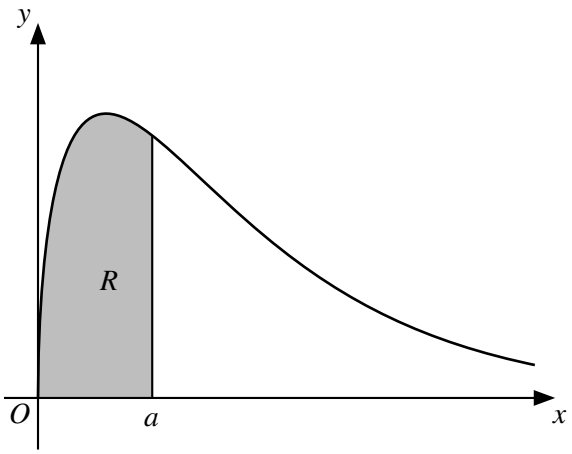
**(a)** Express $f(x)$ in partial fractions. [4]

**(b)** Hence show that

$$\int_0^1 f(x)\,dx = \tfrac{1}{2}\pi - \ln 2.$$ [4]

**[Turn over**

**7**




The diagram shows part of the curve with equation $y = \sqrt{x}\,\mathrm{e}^{-\frac{1}{2}x}$. The region $R$, shaded in the diagram, is bounded by the curve, the $x$-axis and the line $x = a$, where $a$ is a positive constant. A solid ornament is formed by rotating $R$ through 360° about the $x$-axis.

**(a)** Show that the volume of the ornament is $\pi(1 - (a + 1)\mathrm{e}^{-a})$. [4]

**(b)** It is given that the volume of the ornament is $\frac{1}{2}\pi$.

Show that $a = \ln(2a + 2)$. [1]

**(c)** Show by calculation that $a$ lies between 1.5 and 2. [2]

**(d)** Use an iterative formula based on the equation in **7(b)** to find $a$ correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

**8** The complex number $z$ satisfies $|z - 4 - 3\mathrm{i}| = 3$.

**(a)** On an Argand diagram, sketch the locus of the points representing $z$. [2]

**(b)** Find the greatest value of $\arg z$.

Give your answer in the form $\tan^{-1}\left(\frac{p}{q}\right)$, where $p$ and $q$ are integers. [4]

**(c)** Find the complex number $z$ for which $\arg z$ is greatest. Give your answer in the form $x + \mathrm{i}y$, where $x$ and $y$ are exact. [2]

**9** With respect to the origin $O$, the point $A$ has position vector $\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}$.

The line $l$ has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

The point $P$ lies on $l$ and has parameter $\lambda$.

**(a)** Show that $AP^2 = 9\lambda^2 - 18\lambda + 18$. [2]

**(b)** Hence find the shortest distance from $A$ to $l$, and the position vector of the point on $l$ that is nearest to $A$. [3]

**(c)** The points on $l$ whose distance from $A$ is less than 5 form a line segment.

Find the length of this segment. [3]

**Question 10 is on the back page.**

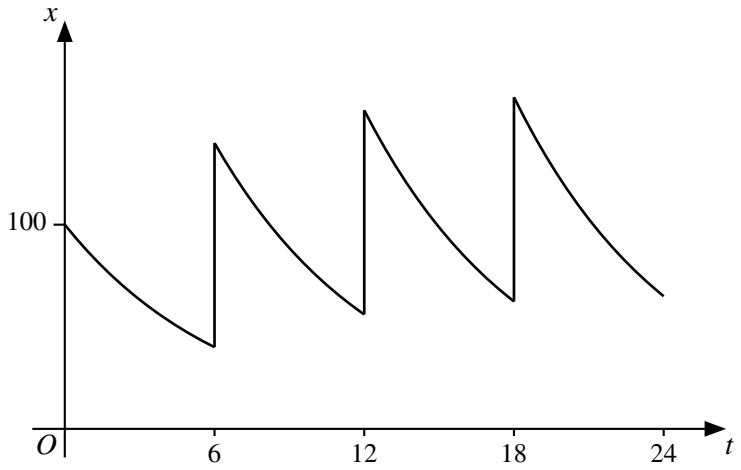
**[Turn over**

**10** A patient is given doses of a drug. At time $t$ hours after the first dose is given, the amount of the drug in the patient's body is $x$ mg. In a model, $x$ is treated as a continuous variable. Between doses, the drug leaves the body at a rate proportional to the amount of the drug in the body.

The first dose is 100 mg and is given at $t = 0$. Just before the second dose is given at $t = 6$, the amount of the drug in the body is 40 mg.

**(a)** Form a differential equation relating $x$ and $t$, for $0 \leqslant t < 6$. Solve it to show that $x = 100 \times 0.4^{t/6}$ for $0 \leqslant t < 6$. [5]

Further doses of 100 mg are given at $t = 6, 12, 18, \ldots$. The diagram shows how $x$ varies with $t$ for the first four doses.




**(b)** Show that, immediately after the $n$th dose, the amount of the drug in the body is $\frac{500}{3}(1 - 0.4^n)$ mg. [3]

**(c)** A doctor decides instead to give a dose of $D$ mg every 6 hours, starting at $t = 0$. The amount of the drug in the patient's body must never be more than 200 mg.

Find the greatest possible value of $D$. [2]